

IMPROVING FEATURE-LEVEL ADVERSARIAL TRANSFERABILITY VIA DIVERSITY ATTRIBUTION

supplementary materials

1. PSEUDOCODE

Algorithm 1 Diversity Attribution Attack

Input: The classifier f , the original image x and its corresponding true label y . **Parameters:** Perturbation magnitude ϵ ; maximum iterations T ; decay factor μ ; ensemble number N ; splitting number s .

- 1: $\alpha = \epsilon/T$; $g_0 = 0$; $x_0^{adv} = x$
 - 2: Obtain N copies of diverse transformations
 - 3: Calculate the diversity attribution:
 - 4:
$$\Delta_k = \frac{1}{C} \sum_{n=1}^N \frac{\partial l(x_n, y)}{\partial f_k(x_n)}$$
 - 5: Construct optimization objective:
 - 6:
$$\mathcal{L}(x^{adv}) = \sum (\Delta_k \odot f_k(x^{adv}))$$
 - 7: Update x^{adv} by momentum iterative method:
 - 8: **for** $t = 0$ to $T - 1$ **do**
 - 9:
$$g_{t+1} = \mu \cdot g_t + \frac{\nabla_x \mathcal{L}(x_t^{adv})}{\|\nabla_x \mathcal{L}(x_t^{adv})\|_1}$$
 - 10:
$$x_{t+1}^{adv} = \text{Clip}_{x, \epsilon} \{x_t^{adv} - \alpha \cdot \text{sign}(g_{t+1})\}$$
 - 11: **end for**
 - 12: **return** $x^{adv} = x_T^{adv}$
-

2. ABLATION STUDIES

2.1. The effectiveness of each transformation method

we further explore the effectiveness of each transformation method. The symbol "+" indicates the introduction of a new transformation method to the image blocks. The initial RPD represents the introduction of only the RPD transformation method to the image blocks. As new transformation methods are gradually added, the diversity of transformations increases, leading to a significant improvement in transferability.

2.2. The parameter settings for the introduced transformation methods

For the parameter setting of the introduced transformation methods, we analyze the optimal parameters for each specific transformation method within the global transformation.

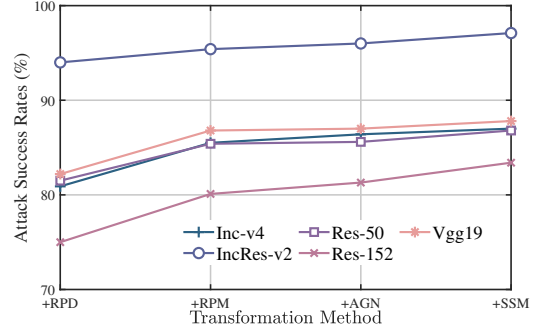


Fig. 1. The effectiveness of each transformation method.

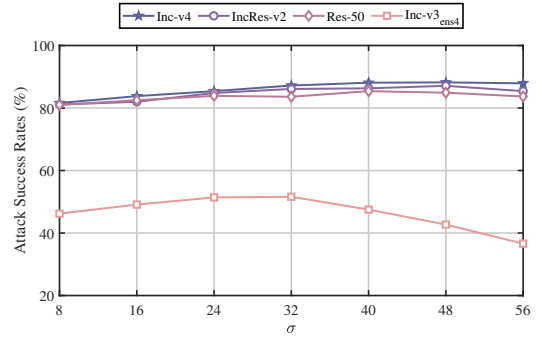


Fig. 2. The hyperparameter σ in the SSM transformation.

The parameter setting of SSM. We analyze the impact of two parameters of the SSM on transferability: the standard deviation σ of Gaussian noise and the transformation factor ρ . As shown in Fig. 2, the attack success rates for the normally trained models peak at $\sigma = 48$. For robust defense models, the peak occurs at $\sigma = 32$ and then begins to decline. Therefore, σ is set to 48 and 32 when attacking normally trained and robust models, respectively. Moreover, For the transformation factor ρ in SSM, as shown in Fig. 3, the attack success rates are higher for normally trained models when $\rho = 0.7$, while $\rho = 0.3$ achieves the best results for robust models. Thus, ρ is set to 0.7 for attacking normally trained models and 0.3 for robust models.

The parameter setting of AGN. We explore the impact

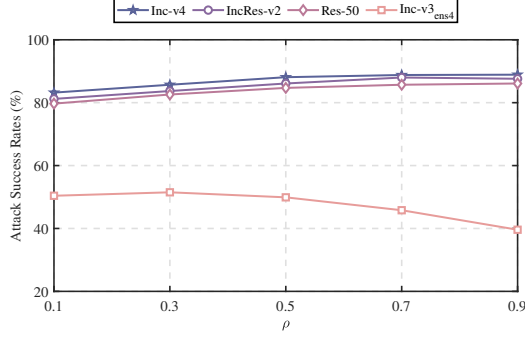


Fig. 3. The hyperparameter ρ in the SSM transformation.

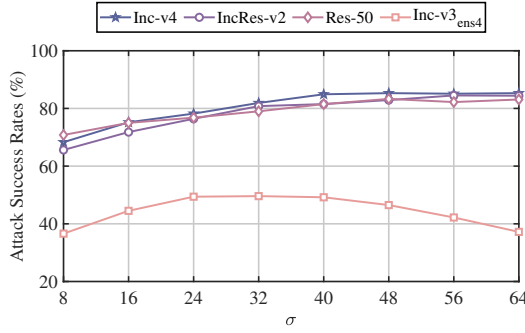


Fig. 4. The hyperparameter σ in the AGN transformation.

of the parameter of the AGN on transferability: the standard deviation σ of Gaussian noise. As shown in Fig. 4, in summary, the attack success rates for normally trained models and robust models reach their maximum at $\sigma = 48$ and $\sigma = 32$. Therefore, σ is set to 48 and 32 when attacking normally trained models and robust models, respectively.

The parameter setting of RPM. Following the settings in [1], the masking probability p_2 is set to 0.3 and 0.2 for attacking normally trained models and robust models, respectively. The patch size is randomly selected from the list [1,3,5,7].

The parameter setting of RPD. Following the settings in [2], the drop probability p_1 is set to 0.3 and 0.1 for attacking normally trained models and robust models, respectively.

3. REFERENCES

- [1] Y. Zhang, Y. Tan, T. Chen, X. Liu, Q. Zhang, and Y. Li, “Enhancing the transferability of adversarial examples with random patch,” in *IJCAI*, 2022, pp. 1672–1678.
- [2] Z. Wang, H. Guo, Z. Zhang, W. Liu, Z. Qin, and K. Ren, “Feature importance-aware transferable adversarial attacks,” in *ICCV*, 2021, pp. 7639–7648.